

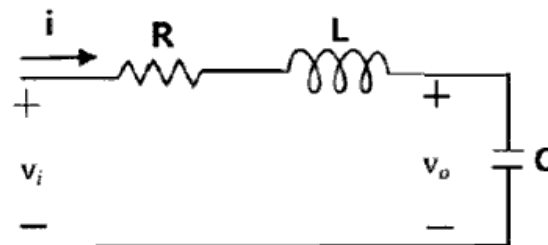
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

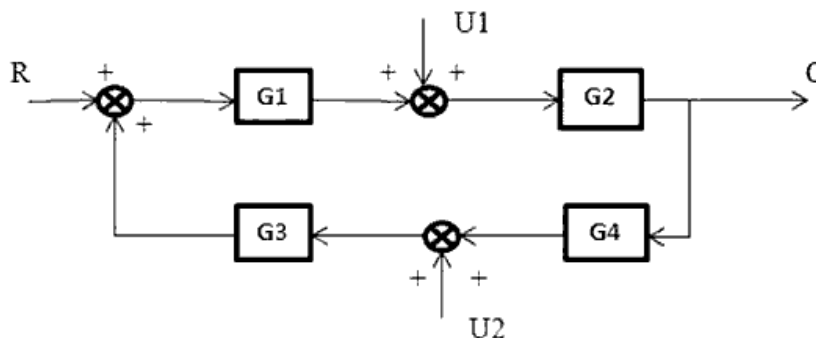
(Level/CO) Marks

Q. 1 Solve Any Two of the following. **12**

- A) Derive Transfer Function Model for the Electrical network shown below **Understand 6**

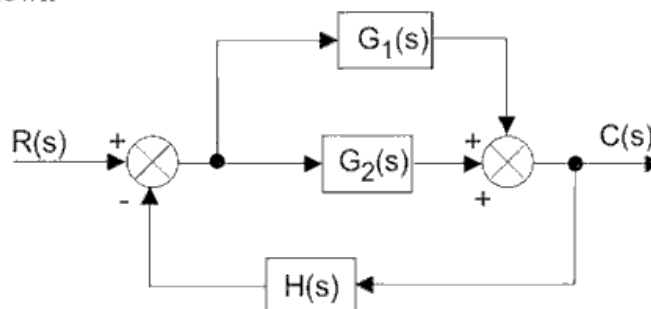


- B) Find the Output in the following block diagram having three inputs R, U1, U2 **Remember 6**



Also find $C/U1$, $C/U2$, C/R

- C) Using Mason's Gain Formula, Find overall transfer function for the Block diagram shown **Remember 6**



Q.2 Solve Any Two of the following. **12**

- A) The Control System Unity Feedback has **Understand 6**

$$G(S) = \frac{20}{S(1+4S)(1+S)}$$

Determine

- i) Type of System
 - ii) Different Static Error Coefficient
 - iii) Steady State error if input, $r(t) = 2 + 4t + \frac{t^2}{2}$
- B) Define Rise Time and Delay time. Also Find the time domain specifications of a control system having the closed loop transfer function **Understand 6**

$$G(S) = \frac{4}{S^2 + 4S + 4}$$

when the unit step signal is applied as an input to this control system.

- C) Explain any two of the following **Understand 6**
- i. Lag Compensator
 - ii. Lead Compensator
 - iii. Lag-Lead Compensator

Q.3 Solve Any Two of the following. **12**

- A) The OLTF of Unity feedback system is given by, **Understand 6**

$$G(S) = \frac{K}{(S+3)(S+5)(S^2+2S+2)}$$

In the Root Locus diagram for this system,

- i) Determine no. of branches of root loci and
 - ii) Determine location at which these branches starts and terminates
 - iii) Find the angle of asymptote and the point at which asymptotes intersect the real axis
 - iv) Find the breakaway point and the point where root loci cross the imaginary axis
- B) Consider the feedback control system with Loop transfer function as , **Understand 6**

$$G(S)H(S) = \frac{1}{(1+S) + (2+S) + (3+S)}$$

Sketch the Polar plot and also check the stability of system.

- C) Using Routh's criterion, Find the stability of the control system having characteristic equation **Remember 6**

$$S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16S + 16 = 0$$

Q.4 Solve Any Two of the following. **12**

- A) Explain the following characteristics of feedback control system **Remember 6**

- Stability
- Disturbance rejection
- Insensitivity

- B) Explain the effect of P, PI, PD control on the performance of control system? **Remember 6**

C) Comment on Stability by drawing Bode Plot For given OLTF .

Understand

6

$$\frac{K(1 + 0.2S)(1 + 0.025S)}{S^3(1 + 0.001S)(1 + 0.005S)}$$

Determine W_{gc} , W_{pc} , G.M. , P.M

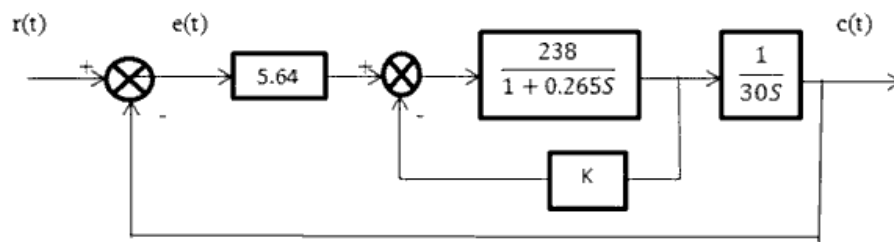
Q. 5 Solve Any Two of the following.

12

A) For Block diagram shown in figure find Natural frequency of system and also find value of K, so that damping is critical.

Understand

6



B) Evaluate Controllability of following system

Understand

6

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} , \quad B = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{And}$$

Evaluate Observability of following system

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} , \quad C = [0 \quad 1]$$

C) Find state transition Matrix of system define by following state model also

Understand

6

find transfer function of the system.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

***** End *****